
Katherine Piper

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Education

Carnegie Mellon University Pittsburgh, Pennsylvania	August 2024 – Present
Doctor of Philosophy Computational Mechanics	Est. April 2029
Master of Science Computational Mechanics	December 2025
California Institute of Technology Pasadena, California	September 2019 – June 2024
Bachelor of Science Mechanical Engineering	

Research Experience

Carnegie Mellon University Pittsburgh, Pennsylvania	
Graduate Researcher	August 2024 – Present
- Constrained optimization machine learning methods for physics and engineering problems	
DEVCOM Army Research Laboratory Aberdeen, Maryland	
Research Associate (Advisors: Jarek Knap and Kevin LoGrande)	Summer 2026
- Advancing Diffusion Maps Surrogates with Constrained Optimization	
NASA Jet Propulsion Laboratory Pasadena, California	
Academic Part Time Employee	October 2023 – August 2024
Year Round Intern	May 2023 – September 2023
- Additive Manufacturing Center (AMC) and Two-Phase Thermal Technology Laboratory	
- Methods led to 100 times cost reduction and 42% performance improvement over state-of-the-art TPL design	
- Two NASA New Technology Reports and US Patent submitted	
California Institute of Technology Pasadena, California	
Summer Undergraduate Research Fellowship	June 2022 – June 2024
Research Advisor – Vinamra Agrawal, Los Alamos National Laboratory	
- Implementation of a model for ductile fracture in functionally graded additively manufactured metals	

Awards

Grand Prize: Sustainable Impact Award:	March 2026
Gresham Smith Innovation Hackathon	
Graduate Fellowship:	May 2024
Carnegie Mellon University College of Engineering	

Research Grants

ACCESS – EXPLORE – National Science Foundation Computing Grant	September 2024 – Present
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Publications and Patents

(Under Review) K. Piper and S. N. Roberts. Beyond Porosity and Permeability: An Alternative Design Space for Passive Capillary Evaporators. Submitted to *Joule*, September 2026.

(Provisional Patent) K. Piper, S. N. Roberts, and T. Daimaru. Novel Geometric Considerations for Solid and Porous Design in Loop Heat Pipe Evaporators. September 2025. *Assigned to California Institute of Technology*

(In Preparation) K. Piper, L. T. Biegler, N. Trask, and K. Dayal. Direct Nonlinear Optimization Enables Reduced Order Models for Contact.

K. Piper and V. Agrawal. Ductile fracture in functionally graded materials: Insight into crack behavior within the gradient interface. arXiv preprint arXiv:2411.18642 (2024)

Media Coverage and Public Engagement

Students pitch sustainability innovations at inaugural Hackathon	March 4, 2026
Carnegie Mellon University Press Release - Project Featured as First Place Team	
Loop Heat Pipes for Smallsat Swarms (LHPss)	October 2024
Scott Roberts: NASA JPL FY24 Progress Report - Named Key Contributor	